



SHALAKA FOUNDATION'S
**KEYSTONE SCHOOL OF
ENGINEERING**



DEPARTMENT OF COMPUTER ENGINEERING

A.Y. 2026-27

Sem-I

Year:TE

Technical Seminar

TECHNICAL SEMINAR TITLE SUBMISSION

1. Seminar Title

Title of Seminar Title: Conceptual Framework of a Digital Twin Fostering Sustainable Manufacturing in a Brownfield Approach of Small Volume Production for SMEs

2. Research Paper Details

- **Authors:** Sandra Krommes, Florian Tomaschko
- **Conference/Book:** Global Conference on Sustainable Manufacturing (GCSM 2022), Springer – Lecture Notes in Mechanical Engineering
- **Year:** 2023
- **DOI:** 10.1007/978-3-031-28839-5_58

3. Introduction

Small and Medium Enterprises (SMEs) often face challenges in adopting Industry 4.0 technologies due to limited resources and high implementation costs. Digital Twin technology creates a virtual model of physical systems, enabling real-time monitoring, simulation, and optimization. This paper presents a conceptual framework for implementing Digital Twin technology in existing manufacturing environments to improve productivity and sustainability.

4. Problem Statement

Many SMEs use outdated manufacturing systems and lack affordable digital solutions for monitoring, predictive maintenance, and process optimization. Implementing Digital Twins in brownfield manufacturing environments is technically challenging and expensive.

5. Objectives

- To understand the concept of Digital Twin technology.
- To study its implementation in SMEs.
- To improve manufacturing efficiency and sustainability.
- To identify challenges and future opportunities in Digital Twin adoption.

6. Proposed Methodology

The paper proposes a Digital Twin framework that integrates physical manufacturing systems with digital models using sensors, Industrial IoT, cloud computing, and data analytics. Real-time data is collected, processed, and analyzed to optimize production and support better decision-making.

7. Technologies Used

- Digital Twin
- Industry 4.0



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- Industrial Internet of Things (IIoT)
- Cloud Computing
- Data Analytics
- Smart Manufacturing

8. Results

The proposed framework improves production monitoring, reduces downtime, enhances resource utilization, supports predictive maintenance, and increases manufacturing sustainability for SMEs.

9. Conclusion

Digital Twin technology has significant potential to transform SMEs by enabling smart manufacturing and sustainable production. The proposed framework provides a practical roadmap for adopting Industry 4.0 technologies in existing manufacturing environments.

10. References

Krommes, S., & Tomaschko, F. (2023). *Conceptual Framework of a Digital Twin Fostering Sustainable Manufacturing in a Brownfield Approach of Small Volume Production for SMEs*. Global Conference on Sustainable Manufacturing (GCSM 2022), Springer. DOI: 10.1007/978-3-031-28839-5_58

11. Student Details

Roll No	Name of Student	PRN
70	Azahar Vajir patel	72431650C

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